

ROADEYE

SMART HELMET SYSTEM

V 1.0

Presented By :

E/21/019 **E/21/371** **E/21/416** **E/21/433**

Every Ride Comes with Invisible Risks

Problems Riders Face

- Blind spots while changing lanes
- Looking down for navigation
- Limited awareness of nearby vehicles
- Delayed emergency assistance
- Weather and road conditions affect riding

Large accident statistics



Current solutions only solve one problem at a time.

None provide an integrated safety solution.

Meet RoadEye



Why RoadEye Wins

Feature	RoadEye	GPS App	Smart Helmet	Dashcam
HUD Navigation	✓	✗	✓	✗
Blind Spot	✓	✗	✗	✗
Crash Detection	✓	✗	Some	✗
Emergency Alerts	✓	✗	Some	✗
Ride Analytics	✓	✗	✗	✗
Cloud Platform	✓	✗	✗	✗

How RoadEye Works

BIKE MODULE

- Proximity Sensing
- IMU
- Environmental Sensors
- ESP32
- SIM Module



HELMET MODULE

- HUD
- Audio Alers
- IMU
- Wear Detection
- TFT Display



MOBILE APP



- Navigation
- Weather
- Ride Analytics
- Emergency Contact Management

HTTPS

CLOUD BACKEND



- User Management
- Rider Storage
- Analytics
- Spring Boot REST API
- Authentication

DATABASE



- Supabase
- SQL
- Auth
- Storage

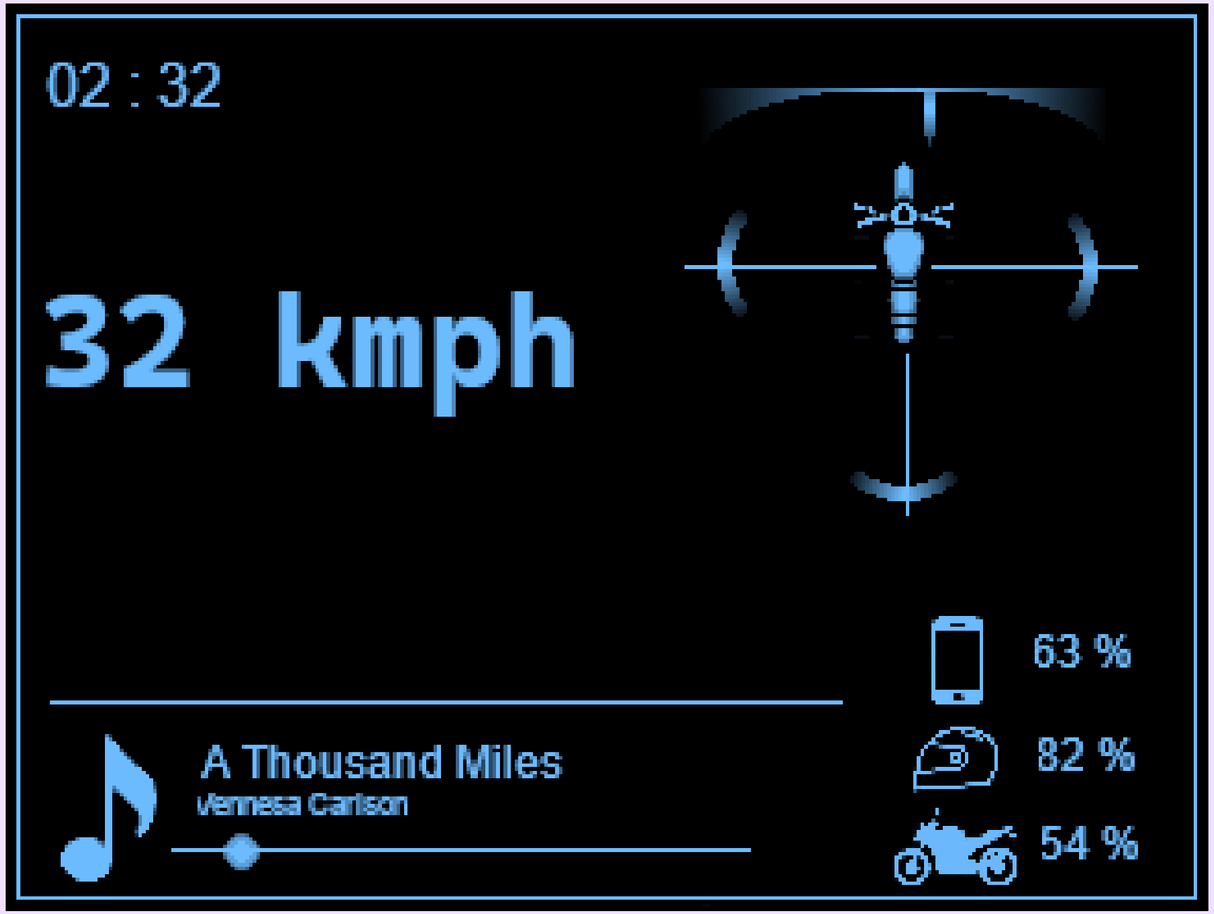
DEMO



✅ **Fully integrated System**

Display Layout





Functional Requirements Satisfaction

Requirements	Status	Requirements	Status
HUD	✓	Mobile Application	✓
Navigation	✓	User Authentication	✓
Crash Detection	✓	Ride Tracking	✓
Emergency Alerts	✓	Cloud Deployment	✓
Proximity Detection	✓	Data Storage	✓

Non-Functional Requirements

DEPENDABILITY

- 99.5% uptime
- Automated backups
- Fault tolerance

EFFICIENCY

- HUD refresh < 60ms
- Alert latency < 200ms

SCALABILITY

- Auto-scaling cloud architecture

SECURITY

- End-to-end encrypted communication

USER EXPERIENCE

- HUD-based interaction
- Voice guidance
- Haptic feedback

JWT Authentication , HTTPS, Secure BLE pairing & Protected user data

Designed for the Road, Not the Desk

Hands-Free First

- Voice guidance and audio alerts
- no touchscreen interaction needed while riding.

Minimal HUD Clutter

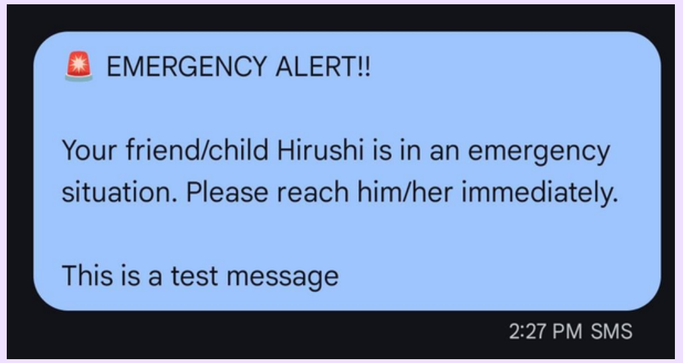
- Only speed, next turn, and alerts appear in view — nothing else competes for attention.

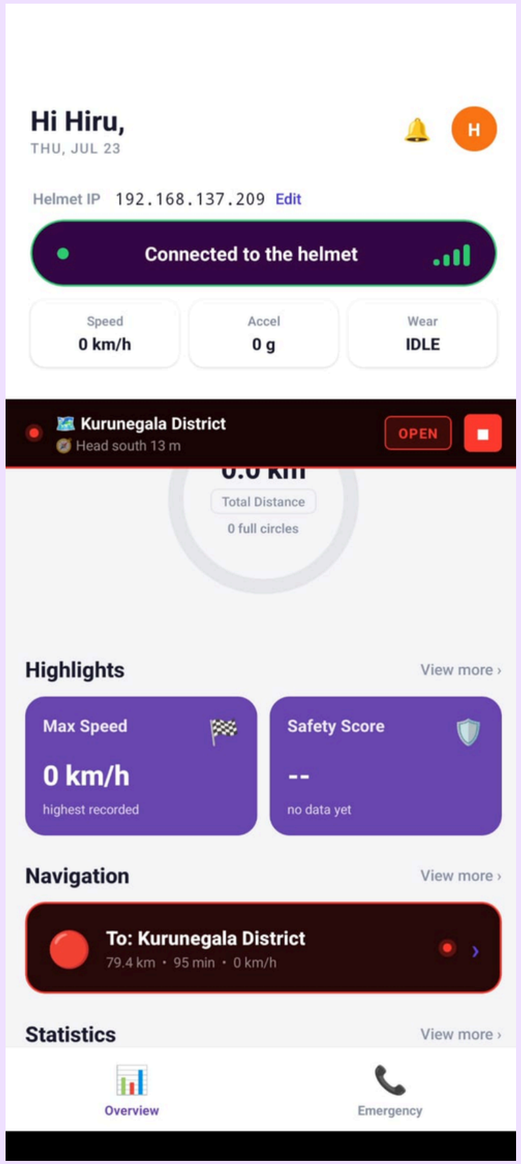
Glanceable Information

- HUD elements sized and placed for a half-second glance, not a stare.

Multi-Sensory Alerts

- Visual + audio + haptic feedback together, so alerts are never missed.





Testing & Validation

Software Testing

Unit Testing



Tools

- JUNit
- Mockito

Integration Testing

- Authentication
- Ride API
- Crash API
- Emergency Notfications



Coverage 85%

Hardware Testing

Sensor Accuracy

MPU6050	±3%
HC-SR04	±2 cm
BME680	±1°C

Tests Conducted

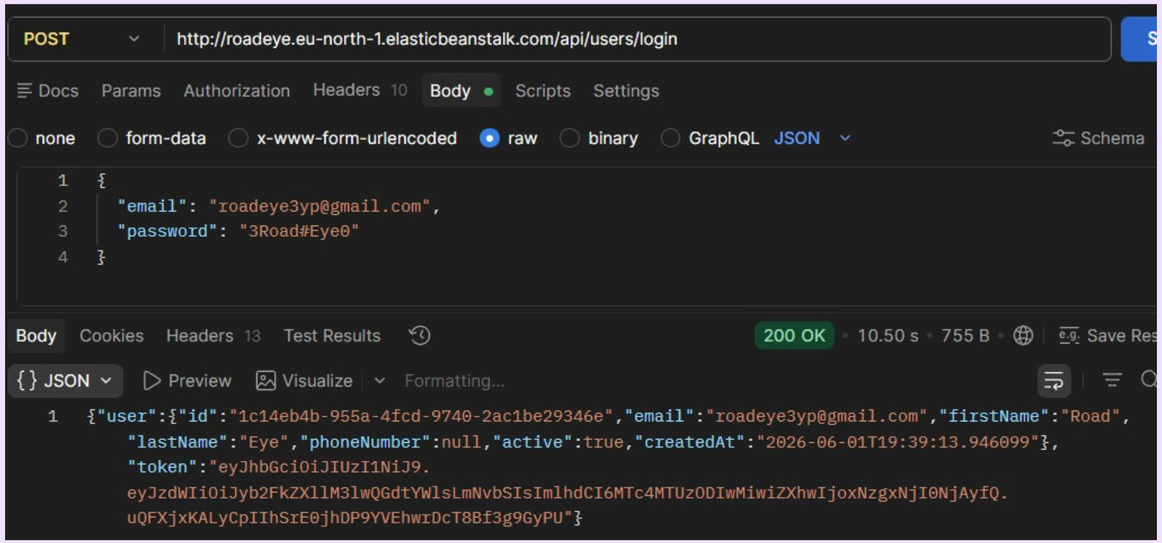
- Crash Detection ✓
- BLE communication ✓
- HUD rendering ✓
- Map Streaming ✓
- Sensor Calibration ✓

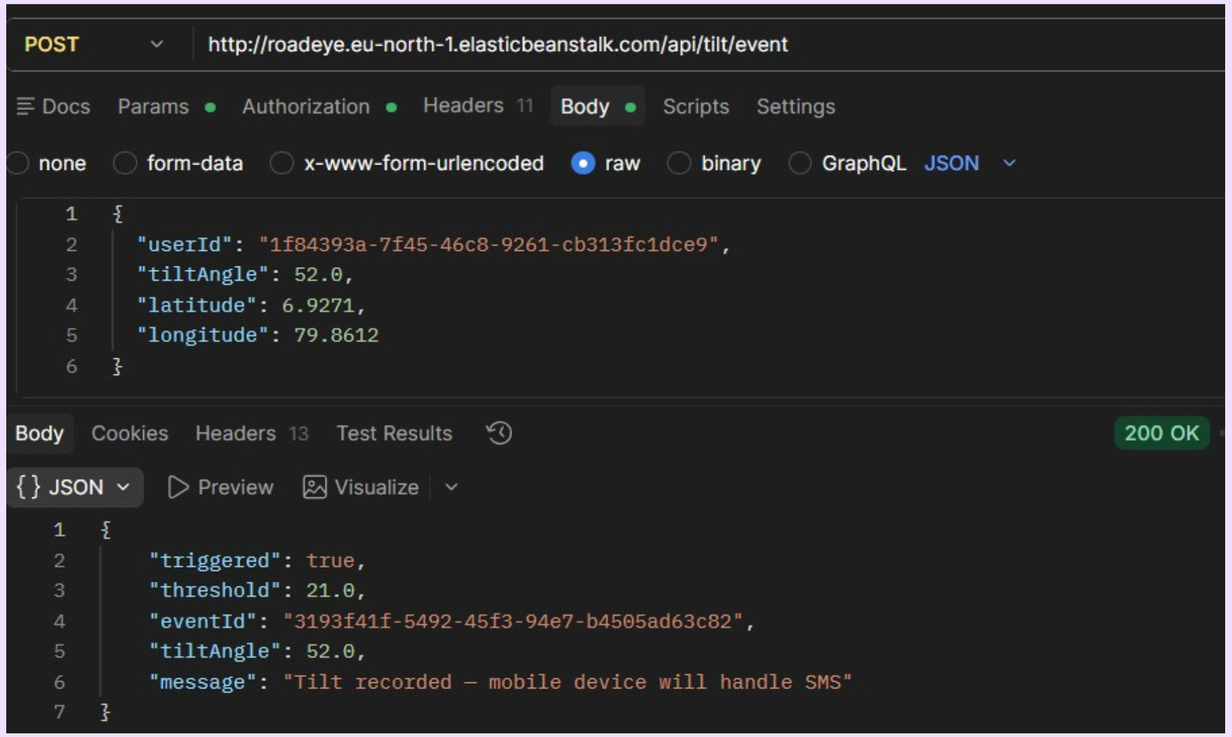
All 28 endpoints and componetns are tested

Testing Proofs - System Intergrated Testing



Testing Proofs - postman





Why RoadEye is Investment Ready

Proven Technology

- Fully integrated hardware ✓
- Mobile application ✓
- Cloud backend ✓
- Live deployment ✓

Growth Potential

- Smart city integration ✓
- AI-powered features ✓
- Helmet-to-Helmet communication ✓
- International expansion ✓

Sustainable Business

- Hardware sales ✓
- Premium subscriptions ✓
- Enterprise solutions ✓

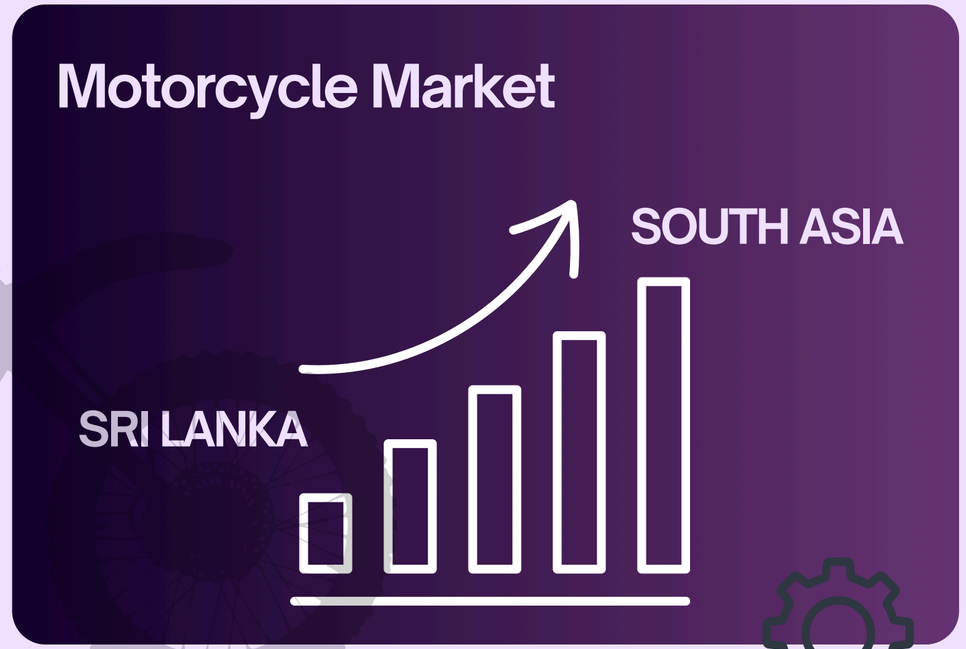
Scalable Platform

- Modular architecture ✓
- Cloud-native infrastructure ✓
- OTA feature updates ✓

Who Needs RoadEye?

- Individual riders
- Delivery riders
- Ride-sharing
- Fleet operators
- Police
- Government
- Insurance companies







Commercialization Strategy

Business Model

Starter Package

- Smart Helmet Module
- Bike Module
- Mobile Application
- Cloud Connectivity
- Navigation & Safety voice Alerts
- Crash Detection
- Ride History

One-Time Hardware Purchase

Premium Subscription

- Automatic Emergency Alert Service
- Advanced Rider Analytics
- Theft Detection
- Extended Cloud Storage
- Multi-device Synchronization
- Priority Customer Support
- AI-powered Riding Insights (Future)

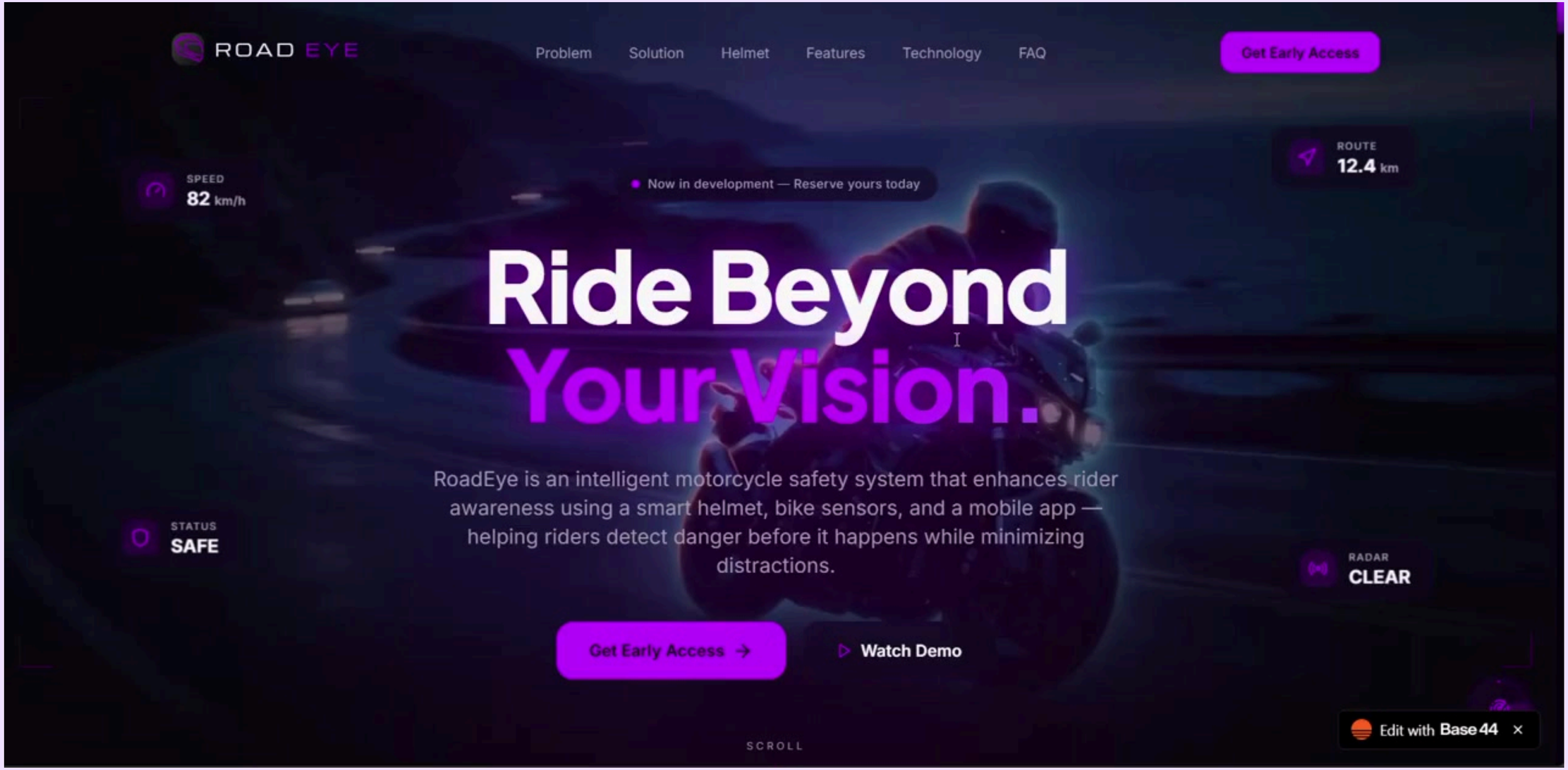
Commercialization Strategy

Enterprise Solutions

- Fleet Management Dashboard
- Delivery Companies
- Ride-sharing Services
- Corporate Motorcycle Fleets
- Insurance Partnerships
- Government Road Safety Programs



RoadEye Ordering Platform



Manufacturing Strategy

Current

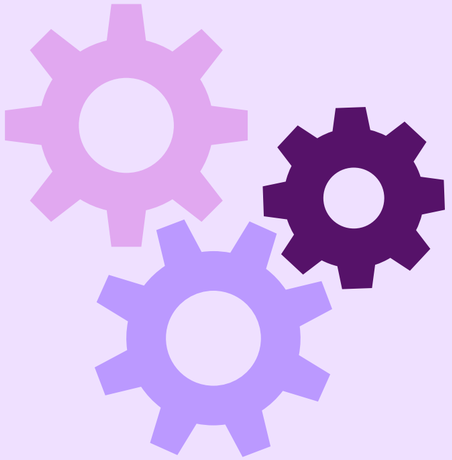
- 3D-printed enclosure
- Off-the-shelf electronics
- Prototype assembly

Next

- Custom PCB
- Injection-molded enclosure
- Waterproof housing
- Product certification

Future

- Manufacturing partner
- Large-scale production
- Reduced unit cost



Growth Roadmap

Current Stage

RoadEye Premium

Connected Riding Platform

- Helmet-to-Helmet Communication
- Group Ride Communication
- AI Hazard Prediction
- Smart Traffic Integration

Commercial Expansion

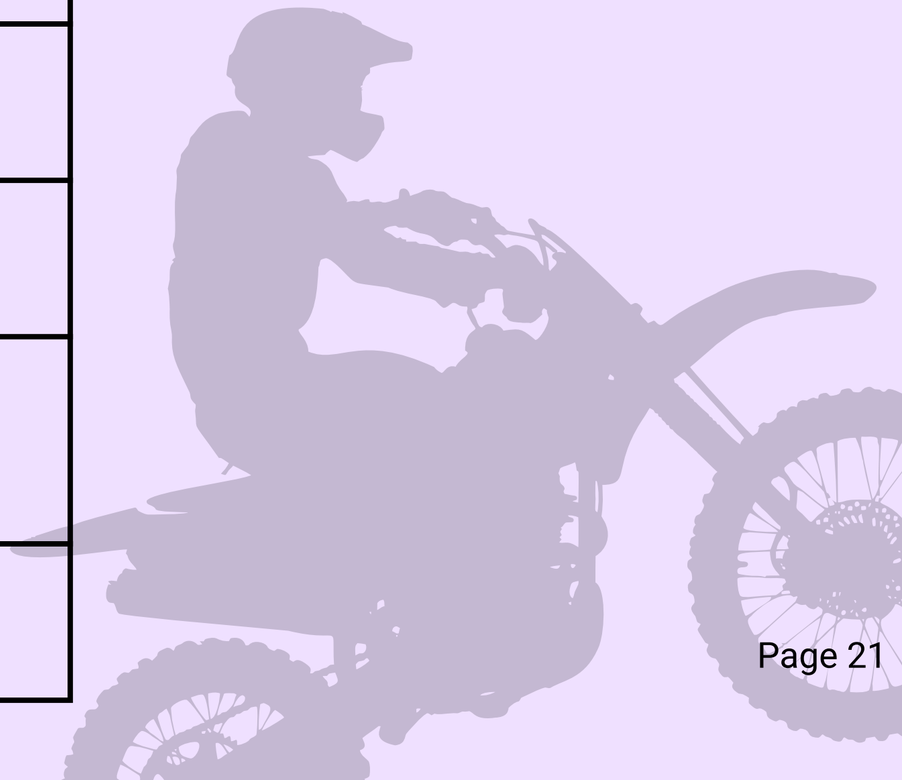
- OEM Partnerships
- Insurance Integration
- Government Road Safety Projects
- International Market Expansion



Total Budget

Module	Price (Rs)
ESP32-C3-MINI-1 microcontroller	1200
Sensors	9800
LiPo and charging circuit , wires	5500
OLED displays	3180
Microphone & Speakers	5800
Fresnel Lens	3500

LKR 28,000



Thank You